

Match the *most appropriate* term to each concept. Not all terms are used and no term is used twice. (2 points each)

Brackish	Hypotonic	Meiofauna
Competitive exclusion	Infauna	Neap tide
Cordgrass	Infralittoral zone	Osmoregulators
Diurnal tide	Interference competition	Salt marsh
Euryhaline	Isotonic	Salt meadow hay
Exploitative competition	Front row at Sea World	Spring tide
Halophile	Littoral	Stenohaline
High marsh	Low marsh	Sublittoral fringe
Hypertonic	Marsh Elder	Supralittoral fringe

1. _____ Tiny organisms that live in the spaces between sand grains.
2. _____ Zone with sufficient light for photosynthesis.
3. _____ Zone between 200–1000 m of depth in the open ocean.
4. _____ Organisms restricted to intermediate salinity concentrations, such as 5–15 ppt.
5. _____ Also known as the splash zone.
6. _____ Organisms that tolerate only a narrow range of high salinity.
7. _____ Free-swimming organisms in water column.
8. _____ All benthic organisms that live *in* the sediment.
9. _____ Strictly marine mammalian herbivore.
10. _____ The open ocean water between the two shelf breaks of an ocean basin.

1 pt extra credit: Another term for #3 above.

(4 pts) 11. Briefly describe how dolphins use high and low frequencies for echolocation.

(5 pts) 12. Briefly explain bioturbation and how it affects the depth of the aerobic zone in a soft-bottom littoral community.

(5 pts) 13. Briefly describe one reason why the size of a tide pool (larger and deeper vs. smaller and shallower) might determine the types of organisms present in the pool.

(10 pts) 14. Compare and contrast a horizontal hard-bottom community to a vertical hard-bottom community in terms of the organisms likely to be present and the important abiotic and biotic factors that determine the differences.

15. Answer the next two questions with regard to a bar-built estuary. Write precise answers and use proper terminology throughout.

(8 pts) (a) Describe the distribution of sediment types in the estuary, and explain why the sediments have this distribution.

(8 pts) (b) Describe the distribution of suspension and deposit feeders in the estuary, and explain why each appears where it does in the estuary.

- (15 pts) 16. Describe and discuss one respiratory and two circulatory adaptations that allow marine cetaceans to dive deep for extended periods of time. Explain how each adaptation enables the cetacean to dive deep for extended periods. Use precise terminology as appropriate.

2 pts extra credit: What are the actual size classes (in mm or μm) for the three sediment types we discussed?

“Great attention gets paid to rainforests because of the diversity of life there. Diversity in the oceans is even greater.”
— Sylvia Earle, Oceanographer.